

Module 1 — Probability

Four modules, one per Revolut topic, from zero upwards: **1 Probability** (this one: rules, random variables, Bernoulli · Binomial · Poisson · Normal) → **2 Statistics** (bias, significance, A/B) → **3 Maths for ML** (bias–variance, validation) → **4 The models** (pros and cons). Every section: the idea in plain words, the formula, a Revolut-shaped example, a 30-second answer, exercises. Answers at the back.

HOW TO USE THIS

Read a section, close the PDF, say the 30-second answer out loud in English, then do the exercises. Only then check the answers. One module per day.

The interview is one hour of *breadth*. The goal is not to be deep anywhere; it is to have a clear, correct, short answer to twenty basic questions and to say "I haven't used that, but I know what it's for" for the rest. Everything here is at that level. Where a topic goes deeper than the round needs, it says so.

00 Reading a formula — ten symbols

SYMBOL	READ IT AS	EXAMPLE
Σ	add up	$\Sigma x_i = x_1 + x_2 + \dots + x_n$ · the mean is $(1/n) \Sigma x_i$
Π	multiply together	probability that several independent things all happen
$P(A)$ · $P(A B)$	probability of A · probability of A given B	$P(\text{default} \text{late last month}) = 0.34$
$E[X]$	expected value: long-run average of X	expected loss per loan
$\text{Var}(X)$ · σ	variance · standard deviation ($\sigma = \sqrt{\text{Var}}$)	how spread out the values are; σ is in the units of X
μ · $\bar{x}$	true (population) mean · sample mean	μ is what you want; $\bar{x}$ is what you can compute
$\log$ · $\exp$	logarithm · its inverse (e^x)	$\log(a \cdot b) = \log a + \log b$ — turns products into sums
$\hat{y}$	"y-hat": the model's prediction of y	y = defaulted (1/0), $\hat{y}$ = predicted probability
$w \cdot x$	dot product = weighted sum	$w_1 x_1 + w_2 x_2 + \dots + w_n x_n$ — every linear model is this
argmin	the value that makes this smallest	training = argmin of the loss over the weights

Probability — rules, random variables, distributions

Three sections: A1 the rules (recap — you own these), A2 random variables, A3 the five distributions. About two hours including exercises.

A1 The rules of probability (you own these — a one-page recap)

- **Probability** = long-run fraction of times something happens. Between 0 and 1. All outcomes sum to 1.
- **AND multiplies:** $P(A \text{ and } B) = P(A) \cdot P(B | A)$. If A and B are independent, $P(B | A) = P(B)$ and it's just $P(A) \cdot P(B)$. "And" always shrinks.
- **OR adds:** $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$. Subtract the overlap so it isn't counted twice.
- **Conditional:** $P(A | B) = P(A \text{ and } B) / P(B)$. "B is now the whole world; what fraction of it is A?"
- **Bayes:** $P(A | B) = P(B | A) \cdot P(A) / P(B)$. Flips the condition. Prior $\times$ likelihood $\div$ evidence = posterior.
- **Complement:** $P(\text{not } A) = 1 - P(A)$. Often the easiest route: $P(\text{at least one}) = 1 - P(\text{none})$.
- **A feature is useful when** $P(\text{target} | \text{feature}) \neq P(\text{target})$. That inequality is what a model learns.

30 SECONDS — "EXPLAIN BAYES' THEOREM"

"It lets you flip a conditional probability. A fraud model that catches 90 % of fraud gives you $P(\text{flag} | \text{fraud})$; the business needs $P(\text{fraud} | \text{flag})$. Bayes multiplies the 90 % by the base rate and divides by how often the model fires overall. With a 0.2 % base rate and a 2 % false-alarm rate, $P(\text{fraud} | \text{flag})$ is about 8 % — most alerts are false, because the rare event is swamped by the small error on the huge legitimate population."

EXERCISES A1

1. Two independent checks each fail 5 % of the time. Probability that at least one fails?
2. $P(A) = 0.3$, $P(B) = 0.5$, $P(A \text{ and } B) = 0.1$. Are A and B independent? What is $P(A | B)$?
3. 1 % of customers are fraudsters. A model flags 80 % of fraudsters and 5 % of honest customers. A customer is flagged — probability they're a fraudster? Do the 10,000-customer table first.

A2 Random variables — a number that depends on chance

A **random variable** is a quantity whose value is decided by a random process: the amount of the next transaction, whether a loan defaults (1 or 0), the number of logins tomorrow. Capital letters for the variable (X), small for a value it takes (x).

Discrete vs. continuous

	DISCRETE	CONTINUOUS
Takes	separate values: 0, 1, 2, ...	any value in a range: 12.37 €
Described by	PMF — probability of each value, $P(X = k)$	PDF — a density curve; probability = area under the curve. $P(X = \text{exactly } 12.37)$ is 0; you ask $P(12 < X < 13)$
Examples	defaults in a portfolio, disputes per month, heads in 10 flips	transaction amount, time until next login, height

Both have **CDF** $F(x) = P(X \leq x)$ — the running total. Percentiles come from it: the median is where $F = 0.5$.
a

Expected value — the long-run average

discrete: $E[X] = \sum x \cdot P(X = x)$ continuous: $E[X] = \int x \cdot f(x) dx$

"Each value, weighted by how likely it is." It is a weighted average, not the most likely value and not the middle value.

REVOLUT

A €1,000 loan: 96 % chance of full repayment (gain €80 interest), 4 % chance of default (lose €600).

$E[\text{profit}] = 0.96 \cdot 80 + 0.04 \cdot (-600) = 76.8 - 24 = €52.8$. Positive → the loan is worth making at that price. Pricing per customer = choosing the interest so that $E[\text{profit}]$ is positive for *that customer's* $P(\text{default})$. That was your lender model.

Two rules of expectation you should know

- **Linearity:** $E[X + Y] = E[X] + E[Y]$, always — even if X and Y are dependent. $E[aX] = a \cdot E[X]$. So the expected loss of a portfolio = sum of the expected losses of the loans. No independence needed.
- **Products:** $E[XY] = E[X] \cdot E[Y]$ only if independent.

Variance and standard deviation — how spread out

$$\text{Var}(X) = E[(X - \mu)^2] = E[X^2] - \mu^2 \quad \sigma = \sqrt{\text{Var}(X)}$$

Variance is the average *squared* distance from the mean (squared so that + and – don't cancel). σ brings it back to the original units, so quote σ . Rules: $\text{Var}(aX) = a^2 \text{Var}(X)$; $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ *only if independent*.

REVOLUT

Two portfolios with the same expected loss of €24 per loan. Portfolio A: every loan loses about €24. Portfolio B: most lose nothing, a few lose €5,000. Same mean, very different variance. Capital requirements, stress tests and "how bad can a bad month be" are all about variance, not the mean.

Standardising: the z-score

$$z = (x - \mu) / \sigma \text{ "how many standard deviations from the mean"}$$

A z of 3 is "unusual" for most well-behaved variables; for a normal variable, only 0.3 % of values are beyond ± 3 . This is the simplest anomaly detector: a customer whose spend today is 8σ above their own mean.

Covariance and correlation — do two variables move together

$$\text{Cov}(X, Y) = E[(X - \mu_x)(Y - \mu_y)] \quad \text{corr}(X, Y) = \text{Cov}(X, Y) / (\sigma_x \sigma_y) \in [-1, +1]$$

Correlation is covariance scaled to be unit-free. $+1$ = perfect straight line up; 0 = no *linear* relation (there can still be a curved one); -1 = perfect line down. Correlation is not causation: ice-cream sales and drownings both rise in summer.

30 SECONDS — "WHAT'S A RANDOM VARIABLE, AND WHAT ARE EXPECTATION AND VARIANCE?"

"A random variable is a number whose value depends on chance — a transaction amount, whether a loan defaults. Its expectation is its long-run average, each value weighted by its probability; its variance is the average squared distance from that mean, and the standard deviation is the square root, in the original units. Expectation tells you what to expect on average; variance tells you how wrong 'on average' can be. In credit, expected loss sets the price; variance sets the capital."

EXERCISES A2

1. A dispute costs €0 with probability 0.7, €20 with probability 0.25, €500 with probability 0.05. $E[\text{cost}]$? Which single value is most likely? Is $E[\text{cost}]$ a value that ever actually occurs?
2. X has mean 10 and σ 2. What are the mean and σ of $3X$? Of $X + 5$?
3. Two independent loans, each with variance of loss 100. Variance of the total loss? What if the two loans were to the same customer (perfectly dependent)?
4. A customer's daily spend: mean €40, σ €15. Today: €130. z-score? Would you call that unusual?
5. Correlation between two features is 0. Can they still be related? Give an example.

A3 The distributions — the five you must know cold

For each: what it models, its parameter(s), mean and variance, the shape, one Revolut example, and when *not* to use it. The interviewer will ask "what is X and when would you use it" — that's the whole question.

1 • Bernoulli — one yes/no trial

- **Models:** a single event that either happens (1) or doesn't (0), with probability p .
- **Parameter:** p . **Mean:** p . **Variance:** $p(1 - p)$ — largest at $p = 0.5$, tiny when p is near 0 or 1.
- **PMF:** $P(X = 1) = p$, $P(X = 0) = 1 - p$.
- **Revolut:** does *this* transaction turn out to be fraud? does *this* customer default? The target of every binary classifier is a Bernoulli variable; a model's output $\hat{y}$ is its estimate of p for that customer.

2 • Binomial — count of successes in n independent trials

- **Models:** the number of "yes" in n independent Bernoulli trials with the same p .
- **Parameters:** n , p . **Mean:** np . **Variance:** $np(1 - p)$.
- **PMF:** $P(X = k) = C(n, k) \cdot p^k \cdot (1 - p)^{n-k}$, where $C(n, k) = n! / (k!(n - k)!)$ counts the ways to choose which k trials succeed.
- **Revolut:** of 1,000 loans with $p = 4\%$, how many default? Mean 40, $\sigma = \sqrt{1000 \cdot 0.04 \cdot 0.96} \approx 6.2$. So 28–52 defaults is "normal"; 70 is not — something changed. Also: an A/B test's conversions per arm are binomial, which is where the A/B formulas come from.
- **Conditions:** fixed n , same p every trial, independence. If loans are correlated (same employer goes bankrupt), binomial *understates* the variance — a classic risk-model error.

WORKED

5 transactions, each 10 % chance of being disputed. $P(\text{exactly 2 disputed}) = C(5, 2) \cdot 0.1^2 \cdot 0.9^3 = 10 \cdot 0.01 \cdot 0.729 = \mathbf{0.073}$. $P(\text{at least one}) = 1 - P(\text{none}) = 1 - 0.9^5 = 1 - 0.590 = \mathbf{0.41}$.

3 • Poisson — count of events in a window, when events are rare and independent

- **Models:** how many events occur in a fixed interval of time or space, when they happen independently at a constant average rate λ .

- **Parameter:** λ (the average count). **Mean:** λ . **Variance:** λ — the same number. That equality is the fingerprint.
- **PMF:** $P(X = k) = e^{(-\lambda)} \cdot \lambda^k / k!$
- **Revolut:** disputes per merchant per month; support tickets per hour; fraud attempts per day on a card. Capacity planning ("how many agents at 3 a.m.?",) is Poisson.
- **Relation to binomial:** binomial with large n and small $p \approx$ Poisson with $\lambda = np$. That's why rare events (fraud among millions) are Poisson-shaped.
- **Diagnostic:** if the observed variance is much larger than the mean (**over-dispersion**), events aren't independent — they cluster: a fraud ring, a broken merchant, a bug. If variance $\ll$ mean, something is smoothing them.

WORKED

A merchant averages 2 disputes a month. $P(0 \text{ this month}) = e^{(-2)} = \mathbf{0.135}$. $P(\geq 5)$ — small: $P(5) = e^{(-2)} \cdot 2^5 / 120 = 0.036$, and the tail beyond adds ~ 0.017 , so $\approx \mathbf{0.05}$. Seeing 8 in a month is a 1-in-500 event $\rightarrow$ investigate.

4 · Normal (Gaussian) — the bell curve

- **Models:** quantities that are the sum or average of many small independent effects. Symmetric, bell-shaped, thin tails.
- **Parameters:** μ (centre), σ (width). **Mean:** μ . **Variance:** σ^2 .
- **PDF:** $f(x) = (1 / (\sigma\sqrt{2\pi})) \cdot \exp(-(x - \mu)^2 / (2\sigma^2))$ — you don't need to compute it; you need the rule below.
- **The 68 / 95 / 99.7 rule:** 68 % of values within $\pm 1\sigma$, 95 % within $\pm 2\sigma$ (precisely 1.96σ), 99.7 % within $\pm 3\sigma$.
- **Standard normal:** $\mu = 0$, $\sigma = 1$. Any normal becomes standard via the z-score. That's why tables and tests work with z .
- **Why it's everywhere:** the **Central Limit Theorem** (Part B): averages of anything become normal as the sample grows. So *sample means*, test statistics and many model errors are normal even when the raw data isn't.
- **Revolut:** the average transaction size of a large sample of customers; the error of a well-behaved regression; a test statistic in an A/B test. **Not** individual transaction amounts, balances, incomes — those are heavy-tailed.
- **How to check normality:** histogram; Q-Q plot (points on the diagonal = normal); mean $\approx$ median; skewness ≈ 0 . Formal tests (Shapiro–Wilk) exist but reject everything at large n — use the plots.

WORKED

Average session length is normal with $\mu = 6$ min, $\sigma = 2$. $P(\text{session} > 10 \text{ min})$? $z = (10 - 6)/2 = 2 \rightarrow$ beyond $+2\sigma$ is 2.5 % (the 95 % rule leaves 5 % in both tails, 2.5 % each). $P(4 < X < 8) =$ within $\pm 1\sigma = \mathbf{68\%}$.

5 · Heavy-tailed / log-normal — money

- **Models:** positive quantities where most values are small and a few are enormous: transaction amounts, balances, income, LTV, city sizes.
- **Log-normal:** X is log-normal if $\log(X)$ is normal. Right-skewed; mean $>$ median; the mean is dragged up by the tail.
- **What to do:** quote the **median and percentiles**, not the mean; take $\log(x)$ before modelling or before a t-test; be suspicious of any "average" of money.
- **Revolut:** "average balance €2,300, median €410" — a few whales. An A/B test on revenue per user is dominated by whales: use a log transform, trimmed means, or a test on the median.

Bonus, one line each

- **Uniform:** every value in $[a, b]$ equally likely. Mean $(a+b)/2$. Random number generators; "no information" priors.
- **Exponential:** time *between* Poisson events; memoryless. Mean $1/\lambda$. Time until the next login, next failure.
- **Geometric:** number of trials until the first success. Expected number of tries until a customer converts.

How they relate — one picture

Bernoulli (one trial) —sum of n —> Binomial (n trials) — n large, p small—> Poisson (rare events, $\lambda = np$) — n large—> Normal (by the CLT) Any variable —average many of them—> Normal (CLT)
Positive & multiplicative —log—> Normal (that's log-normal)

DISTRIBUTION	TYPE	PARAMETERS	MEAN	VARIANCE	USE FOR
Bernoulli	discrete	p	p	$p(1-p)$	one yes/no
Binomial	discrete	n, p	np	$np(1-p)$	count of yes in n trials
Poisson	discrete	λ	λ	λ	count of rare events per window
Normal	continuous	μ, σ	μ	σ^2	sums/averages; errors; test statistics
Log-normal	continuous	μ, σ of $\log X$	> median	large	money, sizes
Uniform	either	a, b	$(a+b)/2$	$(b-a)^2/12$	equally likely values
Exponential	continuous	λ	$1/\lambda$	$1/\lambda^2$	time between events

30 SECONDS — "BERNOULLI, BINOMIAL, POISSON, NORMAL — WHEN DO YOU USE EACH?"

"Bernoulli is one yes/no event with probability p — one transaction being fraud. Binomial counts the yeses in n independent trials — defaults in a portfolio of a thousand loans, mean np . Poisson counts rare, independent events in a time window — disputes per merchant per month, with mean equal to variance, so if the variance is much bigger, events are clustering and I'd look for a ring. Normal is the bell curve: sums and averages of many small effects, so sample means and test statistics are normal by the central limit theorem — but individual amounts of money are not; those are heavy-tailed, and I'd log them or use medians."

EXERCISES A3

1. A card has a 0.3 % chance of a fraud attempt on any given day, days independent. Which distribution for "fraud attempts in a year"? Mean? Which simpler distribution approximates it, and why?
2. 10 loans, each 5 % default, independent. $P(\text{no defaults})$? $P(\text{at least 2})$?
3. Support tickets arrive at 12 per hour. $P(\text{exactly 0 in a 5-minute window})$? (Hint: λ for 5 minutes.)
4. Model residuals are normal with $\sigma = €8$. What fraction of predictions are off by more than €16?
5. You're told "average transaction is €85, standard deviation €600". What does the $\sigma > \text{mean}$ tell you about the shape? Which statistic would you ask for?
6. Disputes per merchant per month: mean 3, variance 30. Poisson or not? What might be going on?
7. Why is the variance of a Bernoulli largest at $p = 0.5$? What does that mean for predicting a 50/50 event vs. a 1 % event?

A Answers — check only after you've written yours

A1

1. $P(\text{at least one fails}) = 1 - P(\text{both pass}) = 1 - 0.95^2 = 1 - 0.9025 = \mathbf{9.75\%}$. (Not 10 %: adding double-counts the case where both fail.)
2. Independent would need $P(A \text{ and } B) = 0.3 \times 0.5 = 0.15$; it's 0.1, so **not independent** (B makes A less likely). $P(A | B) = 0.1 / 0.5 = \mathbf{0.2}$.

3. Per 10,000: 100 fraudsters → 80 flagged; 9,900 honest → 495 flagged. $P(\text{fraud} | \text{flag}) = 80 / 575 = \mathbf{13.9\%}$.
Bayes: $0.8 \cdot 0.01 / (0.8 \cdot 0.01 + 0.05 \cdot 0.99) = 0.008 / 0.0575 = 0.139$.

A2

1. $E = 0.7 \cdot 0 + 0.25 \cdot 20 + 0.05 \cdot 500 = 0 + 5 + 25 = \mathbf{€30}$. Most likely value: €0. And no — €30 never occurs; the expectation is an average, not an outcome. (This is the "mean of money" problem in miniature.)
2. 3X: mean 30, σ 6 (variance scales by 9, σ by 3). $X + 5$: mean 15, σ unchanged at 2 (shifting doesn't change spread).
3. Independent: $100 + 100 = \mathbf{200}$. Perfectly dependent (same customer, both default together): the loss is 2X, so $\text{Var} = 4 \times 100 = \mathbf{400}$. Correlation doubles the risk — that's concentration risk, and it's why binomial models understate portfolio variance when loans are correlated.
4. $z = (130 - 40) / 15 = \mathbf{6}$. Yes, unusual — for a normal variable a 6σ event is one in a billion. But daily spend is heavy-tailed, so "unusual" is weaker than the normal rule suggests; still worth a look.
5. Yes. Correlation only detects *linear* relations. Example: $y = x^2$ over a symmetric range: perfectly related, correlation 0. Or a U-shaped relation between age and default rate.

A3

1. Binomial with $n = 365$, $p = 0.003$. Mean = 1.1 attempts. Because n is large and p small, **Poisson with $\lambda = 1.1$** approximates it well: $P(0) = e^{-(1.1)} \approx 0.33$.
2. $P(\text{none}) = 0.95^{10} = \mathbf{0.599}$. $P(\text{exactly 1}) = 10 \cdot 0.05 \cdot 0.95^9 = 10 \cdot 0.05 \cdot 0.630 = 0.315$. $P(\text{at least 2}) = 1 - 0.599 - 0.315 = \mathbf{0.086}$.
3. λ for 5 minutes = $12 \times 5/60 = 1$. $P(0) = e^{-(1)} = \mathbf{0.368}$.
4. $\mathbf{€16} = 2\sigma$. Beyond $\pm 2\sigma$ is about 5% → **~5%** of predictions off by more than €16.
5. σ much bigger than the mean on a positive quantity means a long right tail — a few huge transactions. The mean is not a typical transaction. Ask for the **median and the 90th/99th percentiles**.
6. Poisson would have variance ≈ 3 . Variance 30 = ten times over-dispersed → **not Poisson**; disputes are clustering — a few merchants (or a few weeks) generate most of them. Look for a fraud ring, a broken merchant integration, or a data bug.
7. $\text{Var} = p(1 - p)$: at $p = 0.5$ it's 0.25, at $p = 0.01$ it's 0.0099. A 50/50 event is maximally uncertain — each observation carries the most information and you need the most data per unit of precision; a 1% event is almost always "no", so each observation tells you little and rare "yes" cases are precious. That's why imbalanced classification is hard: the information is concentrated in the few positives.